
Srikanth Uppiretla

Structural Design and Analysis Engineer

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Status Job seeker visa; **available immediately.**

Portfolio srikanthuppiretla.com

LinkedIn linkedin.com/in/srikanth-uppiretla



Profile

Structural design and analysis engineer with an M.Sc. in Space Engineering and a B.Tech in Mechanical Engineering. At Rocket Factory Augsburg I developed a manufacturable stringer-stiffened CFRP structure and checked it with hand calculations and FEM. I also screened design variants in Python and designed and wound a CFRP shell with ASTRA.

Professional Experience

Apr 2024 - Jun 2025 **Working Student, Structural Analysis (Master's Thesis)** | *Rocket Factory Augsburg AG and DLR, Augsburg and Bremen*
1 year 3 months

- Replaced an expensive isogrid concept with a simpler **stringer-stiffened layout** chosen on **cost and mass**, with cross-sections agreed with the manufacturer.
- Wrote a **Python script** that screened 2,100 skin and stiffener combinations against the NASA SP-8007 and Bruhn buckling standards and found the **lightest design at 165 kg** that met all safety margins.
- Correlated the nonlinear collapse load with the **KDF-corrected linear FE prediction**, achieving a **0.3 % deviation**.
- Built and validated finite element models of a carbon-fiber (CFRP) rocket interstage in **FEMAP and NX Nastran**, covering linear buckling and **nonlinear collapse** analysis.

Jan 2024 - Mar 2024 **Intern, Structural Analysis** | *Rocket Factory Augsburg AG, Augsburg*
3 months

- Sized the **bolted joints** of a nine-engine transport and erection cage across three configurations, carrying the load in friction at 80 % of installed preload with **positive margins in every case**.
- Verified **M8 grade 8.8 joints** under an 1,800 kg engine load at 3.6 g transport and 2 g lateral acceleration, deriving the required preload from FEM beam forces with a 1.25 ultimate load factor.
- Analyzed the **LOX tunnel** under axial and radial pressure, evaluated dome inversion risk and identified stress hotspots for technical reviews.

Aug 2022 - Dec 2023 **Development Engineer (voluntary)** | *ASTRA e.V., student rocketry, Bremen*
1 year 5 months

- Designed, calculated, wound and verified a **filament-wound CFRP outer shell** for a sounding rocket, from laminate definition through manufacture to verification.
- Defined **winding angles, mandrel geometry and ply layup**, translating shell geometry into a laminate that could be wound.
- Ran the **filament winding** on the mandrel, then took the shell through cure and final inspection; refined the laminate with **FE simulations under internal pressure**.

Technical Skills

CAD Software	CATIA V5, Siemens NX, Autodesk Inventor
FEM Software	FEMAP, NX Nastran, ANSYS
Programming & Tools	Python, MATLAB, Git, LaTeX
Methods & Standards	NASA SP-8007, Bruhn, classical laminate theory, GD&T

Education

2020 - 2025	M.Sc. Space Engineering <i>University of Bremen, Germany</i> Key modules: Structural Design and Analysis; Fatigue and Loads; Design of Space Vehicles; Space Systems Engineering.
2013 - 2017	B.Tech Mechanical Engineering (Honours) <i>Lovely Professional University, India</i> Key courses: Design of Machine Elements; Engineering Materials; Computer-Aided Design and Analysis; Machine Tools and Machining.

Further Training

Jul 2025 - today	Professional Development and Engineering Portfolio <i>Self-directed / Udemy</i> - Completed CATIA V5 training (40 h) in Sep 2026; built an engine assembly, planetary gear set and towing assembly with DMU kinematics and derived 2D drawings. - Published a portfolio of four FEA case studies from launch-vehicle and rocketry structural work (srikanthuppiretla.com).
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Selected Project

2017	Four-Wheel Steering Prototype <i>Lovely Professional University</i> Designed and fabricated a working mechanical prototype with front-, rear- and four-wheel steering modes, covering design, assembly and functional checking.
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Languages

English	C1
German	B2
Hindi	Fluent
Telugu	Native